© The Hong Kong Polytechnic University

THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Course : Broad Discipline of COMP-61431, BSc Information Security-42480

Subject : COMP4442 Service and Cloud Computing

Group : 2011, 231, 241

Session : 2022 / 2023 Semester II

Date : 4 May 2023 Time : 12:30 - 14:30

Time Allowed: 2 Hours Subject Lecturer: Dr ZHANG Zhaorui

This question paper has ____5____ pages (cover included).
(Some pages may be intentionally omitted.)

Instructions to Candidates:

- This is an open-book exam.
- The exam is 100 marks in total.
- Any student caught cheating, e.g., sharing or copying answers with classmates, will receive a ZERO mark for the exam.
- Please write your answers in the **Answer Book** and submit it together with the question pages.

Do not turn this page until you are told to do so!

© The Hong Kong Polytechnic University

QUESTION 2 (15 MARKS: TOTAL 100)

Short Questions:

(1) What is the difference between MapReduce and Spark in Hadoop? (3 marks)

(2) What is the Hadoop NameNode and what happens if it fails? (3 marks)

(3) What is the difference between a "transformation" and an "action" in Spark? (3 marks)

(4) What is the role of Spark Executors in a Spark application and how do they work? (3 marks)

(5) Please describe the relationship between Hadoop and Spark. (3 marks)

QUESTION 3 (15 MARKS: TOTAL 100)

Social media applications are very popular and mostly hosted in the cloud. Assume you want to use AWS to build your own social media application which allows users to upload, share, and access their blogs (including photos, texts, and videos) via different end devices. Answer the following questions:

a) Draw the SOA design diagram of your social media application and explain how your design follows SOA principles. (8 marks)

b) Describe the AWS services you would like to use and explain how these services can be orchestrated to implement the functions. (7 marks)

QUESTION 4 (20 MARKS: TOTAL 100)

A hypervisor, also known as a virtual machine monitor (VMM), is a type of software that enables multiple virtual machines to run on a single physical server or host. There are two main types of hypervisor: Type-1 and Type-2. Please answer the following questions:

a) Xen is an open-source hypervisor technology that allows multiple virtual machines to run on a single physical machine, and it runs directly on the host machine's hardware and provides a layer of virtualization that allows multiple operating systems to run independently on the same physical machine.

   According to the above description, which type of virtualization that Xen belongs to, Type-1 or Type-2? (4 marks)

b) Please provide the main difference between Type-1 and Type-2 hypervisors, along with their strengths and weaknesses. (8 marks)

c) Kubernetes is an open-source container orchestration platform that automates the deployment, scaling, and management of the containerized application. It is often used for the management and scheduling of the container launched by Docker, etc. Such virtualization technology is also called containerization, an OS-level virtualization. Please explain what is "OS-Level" virtualization, along with its advantage compared to other virtualization techniques. (8 marks)

© The Hong Kong Polytechnic University

QUESTION 5 (20 MARKS: TOTAL 100)

Healthcare IoT involves using connected devices and sensors to gather data about patients' health status, provide real-time monitoring, and enable better decision-making by healthcare providers. AWS (Amazon Web Services) is a cloud computing platform that provides a wide range of services for IoT applications. AWS IoT offers a variety of features, including secure device connectivity, data collection and analysis, and device management, which are crucial for developing healthcare IoT applications.

a) Analyze what AWS services can be used and how the services can be orchestrated to implement such an application. Draw a diagram and provide a description to illustrate your answer. (10 marks)

b) Following a), suppose you want to extend the system so that the system can use the machine learning algorithm to analyze your health status and provide a report to you. Explain how the designed system in a) can be adapted to support the new function. Draw the new system architecture in a diagram and provide the descriptions. (10 marks)

QUESTION 6 (20 MARKS: TOTAL 100)

A Spark Python program based on RDDs is a program that uses the PySpark API to perform distributed computing tasks on RDDs in Python. The program typically follows a series of steps, including loading data from a data source, transforming the data using RDD operations like map, filter, and reduceByKey, and finally writing the results to an output storage system. Spark Python programs based on RDDs are typically used for processing and analyzing large-scale data sets in a distributed and fault-tolerant manner.

We give a dataset containing a list of temperatures for different cities and dates, and we can write a Spark program in Python based on RDD to find the maximum temperature for each city as follows:

Dataset:

```
City,Date,Temperature
Beijing,2023-03-28,12
Beijing,2023-03-29,16
Hong Kong,2023-03-28,23
Hong Kong,2023-03-29,25
```

Spark Code:

```
from pyspark import SparkContext
sc = SparkContext("local", "MaxTempByCity")
data = sc.textFile("path/to/temperature_data.csv").map(lambda line: line.split(","))
data = data.filter(lambda x: x[1] != "null")
data = data.map(lambda x: (x[0], float(x[1])))
max_temps = data.reduceByKey(lambda x, y: max(x, y))
for city, max_temp in max_temps.collect():
    print(f"{city}: {max_temp}")
sc.stop()
```

© The Hong Kong Polytechnic University

Please answer the following questions based on the above description, dataset and Spark Code:

a) Please provide a description for each line of the code based on the knowledge of Spark and RDD. (6 marks)
b) What is the difference between 'map' function and 'reduceByKey' function in Spark RDDs? How are they used in the context of finding the maximum temperature for each city? (7 marks)
c) What are some advantages and disadvantages of using RDDs compared to DataFrames or other data structures in Spark? When might you choose to use RDDs instead of other structures for temperature data analysis? (7 marks)

END